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Multimodal Correlates of Cannabis Use Among U.S. Veterans with Bipolar Disorder: An Integrated Study of Clinical, Cognitive, and Functional Outcomes

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Abstract

Objective: Cannabis use (CU) is common among persons with bipolar disorder (BD). Evidence suggests that CU is associated with poorer outcomes among persons with BD; however, these findings remain inconsistent. The present exploratory study aims to examine clinical, functional, and cognitive correlates of CU among persons with BD.

Methods: U.S. veterans with BD type I who participated in a large-scale, nationwide study were categorized into four groups: current CU, past CU, past other drug use, and no drug use. Bivariate analyses, univariate analyses of covariance, and Levene's Test for Equality of Variance were used to compare groups on clinical, cognitive, and functional measures.

Results: Of 254 (84.6% male) veterans with BD type I included in the analyses, 13 (5.1%) had current CU, 37 (14.5%) past CU, 77 (30.3%) past other drug users, and 127 (50%) reported no drug use. BD with CU was associated with posttraumatic stress disorder (PTSD) and experiencing lifetime suicidal ideation. Notably, current CU was associated with higher working memory performance, compared to both past CU and no drug use. Likewise, current CU was associated with higher functional capacity, compared to past CU as well as no drug use.

Conclusions: These findings contribute to the growing literature on the complex effects of cannabis on BD. As the commercialization and legalization of cannabis increases, further research in this area is warranted to quantify posed risks to this population, and thereby guide clinical decision-making.

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Keywords

Bipolar disorder; cannabis use; veterans; multimodal correlates

The United States is in the grip of rapid shifts in the legal landscape surrounding the use of cannabinoids, with accompanying changes in public attitudes (Arora et al., 2020; Carliner et al., 2017; Felson et al., 2019; Kilmer & MacCoun, 2017; Schuermeyer et al., 2014). For instance, there is emerging evidence that persons living in states with medical cannabis laws are more likely to use cannabis to relieve symptoms of mood and anxiety disorders (Sarvet et al., 2018). Hence, examining the impact of cannabis use among populations who are likely to use it is of high clinical significance.

Cannabis use (CU) is particularly common among persons with bipolar disorder (BD; Cassidy et al., 2001). Up to 22% of persons with BD report using cannabis at any one time, and up to 64% report having used it in their lifetime, compared with 8.3% (Substance Abuse and Mental Health Services Administration, 2017) and 34%, respectively, among the general population (Sonne et al., 1994). In fact, CU may be second only to alcohol use (AU) among persons with BD (Cassidy et al., 2001; Leweke & Koethe, 2008; Sonne et al., 1994). Further, persons with BD are two to three times more likely than the general population to develop Cannabis Use Disorder (CUD; Koskinen et al., 2010; Stinson et al., 2006).

Several studies suggest a negative impact of CU on the clinical course of BD. For instance, CU among this population has been associated with greater treatment non-compliance (Goldberg et al., 1999; van Rossum et al., 2009; Zorrilla et al., 2015), higher prevalence of psychotic symptoms (van Rossum et al., 2009; Zorrilla et al., 2015), higher incidence of suicide attempts (Dalton et al., 2003; Leite et al., 2015; Weiss et al., 2005; Zorrilla et al., 2015), and a poorer response to lithium treatment (Goldberg et al., 1999) when compared with individuals with BD who do not use cannabis. Conversely, other studies have reported conflicting findings, such as no relationship between current or past cannabis and time to recovery from mood episodes (Ostacher et al., 2010), and that the sequence of onset of CUD and BD may differentially affect outcomes (Strakowski et al., 2007).

The relationship between cognitive symptoms of BD and CU is even less clear. First, converging evidence indicates that a subset of individuals with BD has significant cognitive deficits that are present across mood states. These trait features of BD are associated with worse clinical and functional outcomes, with implications for treatment (Quraishi & Frangou, 2002). Second, among healthy persons, acute CU is associated with deficits in attention, verbal memory, and working memory/executive function, whereas the cognitive effects of *chronic* CU are more complex and remain controversial (D'Souza et al., 2008; Mayor's Committee on Marihuana, 1945; Schoeler et al., 2016; Sewell et al., 2010; Solowij & Pesa, 2010). Third, several studies of the effects of CU on cognition among persons with BD have reported a counterintuitive finding – with persons with BD and CU outperforming persons with BD without CU, particularly on measures of processing speed, attention, and working memory (Braga et al., 2012). Conversely, results from recent studies support the notion that persons with BD who use cannabis have greater cognitive *deficits* than persons with BD without CU (Halder et al., 2016; Vishwakarma et al., n.d.). However, along with limitations such as small sample sizes, these studies also lacked clinical assessments, functional assessments, or both (Braga et al., 2012; Ringen et al., 2010; Sagar et al., 2016). Taken together, while cognitive deficits are well recognized in BD and cannabis may impair cognition in healthy persons, the directionality of the relationship of CU and cognition in BD remains controversial. Further, the relationship between CU and functional outcomes in BD – such as the ability to perform everyday communication and finance-related tasks – has not yet been investigated.

Given the paucity of research in this area, one approach is to examine the relationship between CU and BD using large, existing albeit observational datasets. The Veteran Affairs (VA) Cooperative Studies Program (CSP) 572, a multisite observational study that examined the genetics of functional disability among persons with serious mental illnesses, offers such an opportunity. Not only U.S. veterans are more likely than the general population to use cannabis (Wagner et al., 2007), but veterans with BD are more likely to have posttraumatic stress disorder (PTSD; Thatcher et al., 2007) and experience suicidality (Kvitland et al., 2016; Leite et al., 2015). Taken together, these characteristics may influence the clinical, cognitive, and functional response to cannabis and its constituent cannabinoids. A total of 8,140 participants with BD enrolled in this study, out of whom 270 were enrolled at the West Haven VA site of CSP # 572 study, allowing us to identify participants with BD with and without CU. The study included clinical assessments of co-morbidities, such as suicidality and PTSD, as well as extensive cognitive and functional assessments. Hence, in this exploratory study, we sought to examine clinical, functional, and cognitive correlates of CU among veterans with BD.

Methods

Sample

The study was conducted in accordance with the Declaration of Helsinki and approved by the West Haven VA Human Subjects System and the Yale School of Medicine Human Investigation Committee. All participants in the West Haven site of the CSP 572 study who were diagnosed with BD type I were considered for inclusion in the current study. For inclusion, participants were required to have: 1) a DSM-IV diagnosis of BD type I, determined by the Structured Clinical Interview for DSM-IV (SCID-IV), and 2) an assessment of substance use, available in CSP 572 source documents. We identified participants with and without CU by examining assessments of substance use. A total of 263 participants were enrolled and 254 included in the analyses, due to missing data on 9 participants. The participants were evaluated between January 1, 2014 and December 31, 2015. The present study was conducted using only de-identified data.

Clinical, Cognitive, And Functional Assessments

A detailed description of the assessments is included in the Supplemental Material and in the CSP 572 Methods Paper (Harvey et al., 2014). Clinical assessments included psychiatric diagnoses, suicidality, and data gathering of current pharmacotherapies (Harvey et al., 2014).

The cognitive assessments included 1) The Trail-making part A (attention and psychomotor speed; Strauss et al., 2006), 2) The Brief Assessment of Cognition in Schizophrenia (BACS) Symbol coding subtest (attention and speed of information processing; Keefe, 2004), 3) the Hopkins Verbal Learning Test (HVLT) (episodic memory; Benedict et al., 1998; Brandt, 1991), 4) animal naming (i.e., verbal fluency; Tombaugh, 1999), 5) Letter-Number Sequencing (LNS; for working memory; Wechsler, 2008), and 6) the Neuropsychological Assessment Battery (NAB) mazes test (executive function; Gavett, 2011). All these tests constitute the primary domains of functioning identified in the Measurement and Treatment Research to Improve Cognition in Schizophrenia (MATRICS) Cognitive Consensus Battery (MCCB), which is believed to adequately capture the overall level of impairment in cognitive functioning among persons with serious mental illnesses (Keefe et al., 2006).

Two different performance-based functional capacity measures were administered: 1) The University of California San Diego Performance-based Skills Assessment-B (UPSA-B); and 2) the Everyday Functioning Battery (EFB), both of which have previous evidence of high psychometric quality for everyday functional disability in patients with BD (Harvey et al., 2011, 2013; Keefe et al., 2011). Participants' functional abilities were assessed using the UPSA-B (Mausbach et al., 2007), a measure of functional capacity in which patients are asked to perform everyday tasks related to communication and finances. These instruments were selected because it measures abilities are considered important for independent living (Harvey et al., 2014).

Statistical Analyses

Persons with BD were categorized into 4 groups for comparison: current cannabis use, past cannabis use, past other drug use, and no drug use. First, we used Chi-square analyses to compare the four groups on categorical demographic variables – including sex, race, and employment status – as well as categorical clinical variables, such as PTSD, cigarette use, alcohol use, use of pharmacotherapies, and reported lifetime suicidal thoughts. Previous research indicates that pharmacotherapies can influence the cognitive performance among persons with BD (Arts et al., 2013; Pfennig et al., 2014; Pigoni et al., 2020). Further, we used independent *t*-tests to compare the groups with regards to continuous variables, such as age and Columbia-Suicide Severity Rating Scale (CSSRS) suicidal ideation subscale scores. Variables that significantly differed between groups were used as covariates in subsequent analyses. Next, we used univariate analyses of covariance (ANCOVA) to compare the four groups on clinical, cognitive, and functional measures. All cognitive variables were normally distributed, and Levene's Test for Equality of Variance revealed no significant group differences; therefore, raw data scores from each measure were used for subsequent analyses. *P*-values were FDR-adjusted for multiple comparisons (Glickman et al., 2014).

Results

A total of 263 veterans with BD type I were enrolled in the study, out of whom 254 were included in the analyses; 13 reported current CU, 37 past CU, 77 past other drug use, and 127 no drug use.

Demographics

Veterans with BD and *past* CU were younger than those with BD and past other drug use, with a mean difference of −2.06 (95% CI [−6.38, 2.25], t[111] = −0.948, p = .014). There were no other statistically significant differences in age distributions. Veterans with BD and *current* CU group were more likely to identify as Black or African American (37.1%), compared to those with BD and no drug use, (5.9%; χ^2[3] = 23.51, p < .001). There were no other statistically significant differences in race distributions (Table 1).

Clinical Correlates

Veterans with BD and *current* CU were more likely to have PTSD (46.2%), compared to those with BD and *past* CU (16.7%; χ^2[1] = 4.49, p = .036), as well as persons with BD and no drug use (15.7%; χ^2[1] = 7.21, p = .016).

Veterans with BD and *past* CU were more likely to have experienced suicidal thoughts (75%), compared to those with BD and no drug use (55.1%; χ^2[1] = 4.60, p = .04). Veterans with BD and *past* CU were more likely to have a self-injury attempt (27.8%), compared to those with BD and no drug use (11.1%; χ^2[1] = 6.16, p = .026), as well as compared to those with BD and past other drug use (12.3%; χ^2[1] = 3.99, p = .046).

Veterans with BD and *current* CU were more likely to have an interrupted self-injury attempt (46.2%), compared to those with BD and no drug use (12.6%; χ^2[1] = 10.02, p = .014). Similarly, veterans with BD and *past* CU were more likely to have an interrupted self-injury attempt (33.3%), compared to those with BD and no drug use (12.6%; χ^2[1] = 8.47, p = .014). Veterans with BD and past CU were also more likely to have reported an aborted suicide attempt (33.3%), compared to veterans with BD and no drug use (15.7%; χ^2[1] = 5.49, p = .029).

Finally, there were no statistically significant differences in pharmacotherapy classes. Clinical characteristics are listed in Table 2.

Cognitive Correlates

Persons with BD and *past* CU had a poorer working memory performance, compared to persons with BD and no drug use, indexed by the Mazes test (F[1, 152] = 8.492, p = .011), and the LNS test (F[1, 153] = 6.067, p = .026). These results remained statistically significant when controlled for variables that differed between these groups – namely, reported lifetime suicidal thoughts, race, cigarette use, alcohol abuse, and other medication used as covariates.

Persons with BD and current CU had a better working memory performance than persons with BD and past CU, indexed by the LNS test (F[1, 46] = 4.834, p = .038). These results remained statistically significant after controlling for PTSD diagnosis. There were no other statistically significant differences in cognitive correlates.

Functional Correlates

Persons with BD and current CU had better performance-based functional capacity, compared to persons with BD and past CU (F[1, 46] = 5.496, p = .032). These findings remained significant when controlled for PTSD diagnosis. Cognitive and functional data are shown in z-score scale, with a mean of zero and a standard deviation of one (Figure 1).

Discussion

This is the first study, to our knowledge, to simultaneously explore clinical, cognitive, and functional correlates of CU among veterans with BD. Results from our analyses indicate that: First, CU among veterans with BD was associated with having experienced suicidal thoughts, as well as with PTSD; second, *current* CU was associated with higher working memory performance compared to *past* CU and no drug use; and third, *current* CU was associated with higher functional capacity compared to *past* C, as well as no drug use. Collectively, these findings underscore the complex relationship between dynamics of CU and the phenomenology of BD.

Mounting evidence suggests that CU may influence the expression of suicidality among persons with BD. Our findings are consistent with prior evidence suggesting an association between CU and increased suicidality among persons with BD (Bartoli et al., 2019). Such association may have several possible explanations. First, CU may directly increase risk of persons with BD by precipitating an early onset and potentially increasing the severity of affective episodes (Pinto et al., 2019) – with more severe impulsivity, poor medication adherence, higher prevalence of mixed affective states, and rapid cycling (Foglia et al., 2017; Hawton et al., 2005; Lagerberg et al., 2016; Sublette et al., 2009). Taken together, these factors may exacerbate suicidality. Second, CU is associated with other clinical characteristics that are also known to indirectly increase the risk of suicidal thoughts and related behavior, such as alcohol and other substance misuse, in addition to personality disorders (Carrà et al., 2014; Jylhä et al., 2016).

We also found that persons with BD and CU were more likely to have PTSD. This finding suggests a more severe clinical implication for persons with BD and CU compared to persons with BD and no drug use. Data from the National Veteran’s Association reported by Bowe and Rosenheck shows that veterans with comorbid substance use disorder and PTSD showed to have higher rates of BD (Bowe & Rosenheck, 2015). However, our findings that indicate a specific association between PTSD and CU among persons with BD may be explained by several lines of evidence. First, persons with BD often report using cannabis to cope with mood-related symptoms (Grinspoon & Bakalar, 1998; Gruber et al., 1996; Healey et al., 2009). Second, preclinical and human pharmacological studies provide some biological plausibility to the hypothesis that self-medication of acute anxiety may partly explain the high rates of CU among individuals with PTSD (Bedse et al., 2017; Bluett et al., 2017; Cougle et al., 2011). Further, results from Ringen revealed that persons with BD who reported cannabis use in the past 6 months had significantly lower ratings of anxiety compared to non-users, as well as better scores on neurocognitive tests, such as focused attention, executive functioning, memory learning and recall (Ringen et al., 2010). Still, these findings are associational and warrant further investigation in longitudinal studies.

Moreover, cannabis may *acutely* reduce anxiety, especially at low doses/potencies; *chronic* CU may cause downregulation of cannabinoid 1 receptors in the brain (Hirvonen et al., 2012), increasing the neuroendocrine response to stress (Rey et al., 2012). *Chronic* cannabis use may impact processes that are relevant to the phenomenology of PTSD, such as fear extinction, thereby potentially perpetuating this condition among persons with BD (Papini et al., 2017).

In line with accumulating evidence, we found that CU was associated with superior performance in working memory tasks among persons with BD. This is consistent with prior studies that suggest persons with BD and CU had a better cognitive performance on measures of attention, processing speed, working memory (Braga et al., 2012), and verbal fluency (Ringen et al., 2010). Since CU acutely impairs working memory performance and since chronic CU does not have chronic precognitive effects among other populations, it is unlikely that CU directly enhances cognitive performance among persons with BD (De Aquino et al., 2018; D'Souza et al., 2004). Rather, perhaps CU has second-order effects, for instance, by reducing anxiety, indirectly enhancing task-based cognitive performance; or that superior cognitive performance is a pre-existing characteristic of persons with BD who tend to use cannabis. Notably, in accordance with the latter hypothesis, multimodality biomarker phenotyping studies have shown that, compared with persons with BD and no CU, persons with BD and CU were less impaired in cognitive measures (Clementz et al., 2016). Moreover, emerging evidence from neuroimaging studies indicate that persons with early psychosis (<5 years illness) who use cannabis may have preserved cortical thickness in brain regions related to cognitive performance in working memory tasks (Reed et al., 2021).

That persons with BD and CU had higher functional capacity is also in accordance with some prior research. In a large prospective study, persons with BD and CU engaged in more social activities than persons with BD and no CU (van Rossum et al., 2009). Although this hypothesis requires systematic investigation, the act of buying substances may require cognitive planning, execution, and social interaction. Since CU is associated with several negative clinical outcomes that are known to impair functioning – e.g., frequent mood episodes (Gibbs et al., 2015; Lagerberg et al., 2016; Strakowski et al., 2007), higher risk of hospitalization (Lagerberg et al., 2016), and suicidal ideation (Leite et al., 2015) – the superior functional capacity is likely upstream of CU among persons with BD, rather than a direct effect of CU. This hypothesis is strengthened by the growing data underscoring poorer clinical outcomes, which are generally associated with lower functioning. Alternatively, it is possible that short-term anxiolytic effects of CU may temporarily enhance functional capacity, especially among persons with BD who have functionally impairing dysphoria or anxiety. To the contrary, a large study examining functional outcomes in BD found that females with lifetime CU had poorer functioning among domains such as sexual, financial, social, and emotional functionality compared to females without a history of CU (de la Fuente-Tomás, et al., 2020). Hence, the relationship between CU and functional capacity measures we encountered requires further investigation.

Limitations

This study has several noteworthy strengths, such as being the first to simultaneously characterize the clinical, cognitive, and functional correlates of CU among persons with BD. Further, this is the first study to examine the relationship between CU and BD among U.S. veterans.

Despite notable strengths, there are also limitations to consider. First, the CU data is based on retrospective self-report. A granular estimation CU in terms of amount and frequency was unavailable for this post-hoc analyses. Second, we acknowledge that a precise definition of *past* CU is imperfect. Still, evidence indicates that the cognitive effects of cannabis are time-dependent and may largely resolve after a few weeks of abstinence (Schreiner & Dunn, 2012); thus, and parsing it from *current* CU, allowed us to conduct exploratory analyses and generate hypotheses considering the time-dependent effect of cannabis, which should be specifically addressed in future studies.

Finally, residual confounding due to alcohol use is a possibility. Although we have adjusted for multiple confounding variables, studies examining the relationship between cognitive performance and cannabis use have recognized that residual confounding may remain, secondary to alcohol use (Hall & Degenhardt, 2014; McKetin et al., 2016; Pope & Yurgelun-Todd, 1996).

Implications and Future Directions

Some methodological, mechanistic, and clinical implications can be drawn from our findings. Methodologically, longitudinal studies are necessary, preferably using Ecological Momentary Assessment (EMA). EMA studies proximal data collection and multimodal assessment of mood, cognition, and functional outcomes in real-time and real-world settings. Thus, EMA studies are poised to better characterize the temporal dynamics of CU (acute vs. chronic use), other substance use (e.g., alcohol), and BD symptomatology.

Mechanistically, differential effects of cannabis on various aspects of the BD syndrome – mood, cognition, and functioning – must be considered. In a recent study conducted by the Bipolar Schizophrenia Network of Intermediate Phenotypes (B-SNIP), participants with Early Psychosis and CU had preserved cortical thickness in brain areas related to attention and language (Reed et al., 2021). There is also a growing body of research on the role of the Endocannabinoid System (ECS) in the pathophysiology of BD (Arjmand et al., 2019).

Clinically, growing findings of superior cognitive functioning among persons with BD and CU highlight the promise of pharmacotherapy agents with similar properties as that of cannabinoids, but without the associated hedonistic or psychotomimetic effects, to be used as therapeutics for the cognitive dysfunction associated with BD.

Conclusions

BD is a dynamic syndrome with three affective phases, and the effects of cannabinoids are equally complex. This study elicits an elaborate pattern of superior cognitive performance and functional capacity, despite negative clinical correlates, among veterans with BD who use cannabis. In the absence of experimental data, these analyses contribute to the growing literature on the relationship between CU and BD. There is an urgent need for longitudinal studies with more granular assessments of CU, such as EMA studies, in addition to pharmacological human laboratory studies, and clinical trials. Against the backdrop of rapidly changing attitudes towards cannabis, a convergence of experimental and observational approaches is necessary for characterizing the dynamics of CU and symptoms of BD, as well as the underlying neurobiology. Altogether, such studies will be able to guide decision-making for persons with BD, healthcare providers, and policy makers.

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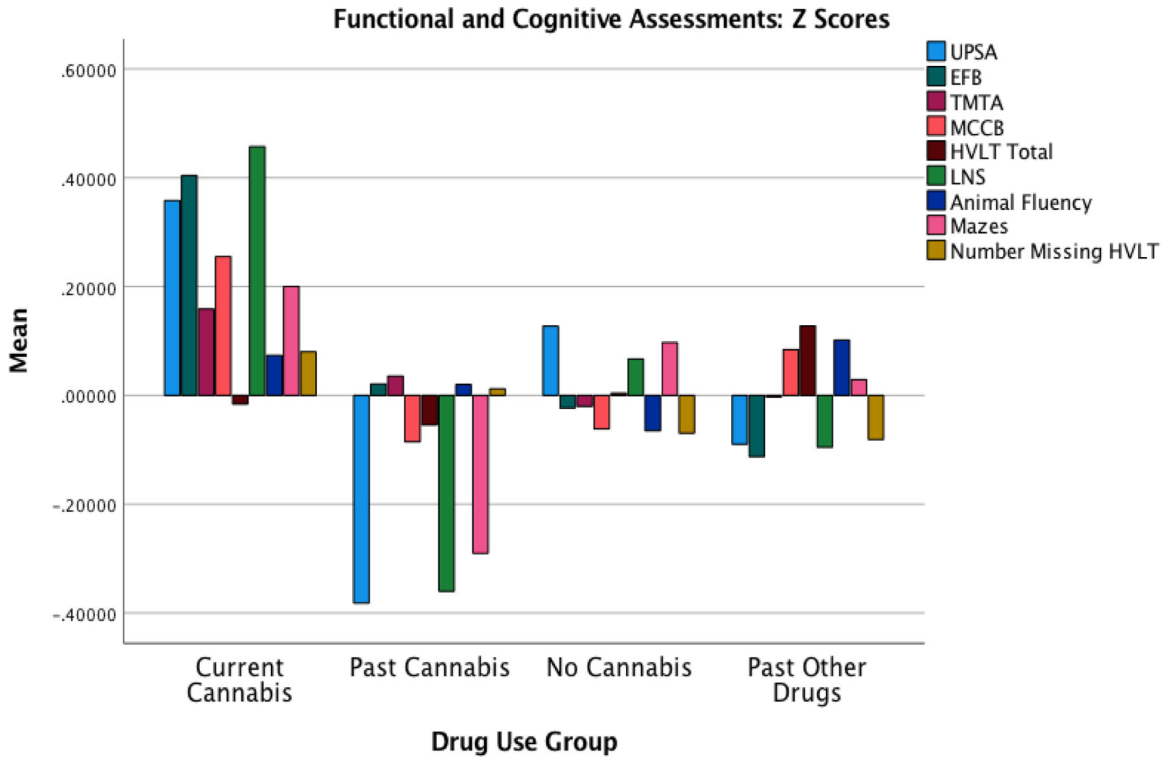

Figure 1.

Functional and Cognitive Performance by Drug Use Group

Note. Persons with past Cannabis Use (CU) had significantly poorer performance on working memory tasks (Mazes; ($F(1, 152) = 8.492$, $p = .011$), (LNS; ($F(1, 153) = 6.067$, $p = .026$). Persons with current CU had significantly better working memory performance than persons with past CU, (LNS; ($F(1, 46) = 4.834$, $p = .038$). Persons with current CU had better performance-based functional capacity, compared to persons with past CU ($F(1, 46) = 5.496$, $p = .032$).

Table 1

Demographics

		Drug Use Group											
		Current Cannabis			Past Cannabis			No Drug Use			Past Other Drugs		
		Count	*N* %	Mean	Count	*N* %	Mean	Count	*N* %	Mean	Count	*N* %	Mean
Sex	**Male**	10	76.9%		31	83.8%		107	84.3%		67	90.5%	
	Female	3	23.1%		6	16.2%		20	15.7%		7	9.5%	
Race	**White**	11	84.6%		22	59.5%		111	87.4%		58	75.3%	
	Black	2	15.4%		13	35.1%		7	5.5%		16	20.8%	
	Asian	0	0.0%		0	0.0%		2	1.6%		0	0.0%	
	Hispanic	0	0.0%		1	2.7%		4	3.1%		3	3.9%	
	Multiracial	0	0.0%		1	2.7%		1	0.8%		0	0.0%	
	Other	0	0.0%		0	0.0%		2	1.6%		0	0.0%	
	Age			50			53			55			51

Table 2

Clinical Characteristics

	Drug Use Group							
	Current Cannabis		Past Cannabis		No Drug Use		Past Other Drugs	
	Count	*N* %	Count	*N* %	Count	*N* %	Count	*N* %
PTSD[•*†*]	6	46.2%	6	16.7%	20	15.7%	17	23.3%
Suicidal Thoughts[‡*]	8	61.5%	27	75.0%	70	55.1%	48	64.9%
Self-Injury Attempt[‡*¤*]	2	15.4%	10	27.8%	14	11.1%	9	12.3%
Interrupted Self-Injury Attempt[†*‡*]	6	46.2%	12	33.3%	16	12.6%	15	20.3%
Aborted Suicide Attempt[‡*]	1	8.3%	12	33.3%	20	15.7%	19	25.7%
Benzodiazepine	0	0.0%	6	16.2%	28	22.0%	16	20.8%
Antidepressant	5	38.5%	12	32.4%	55	43.3%	32	41.6%
Antipsychotic[‖*]	5	38.5%	16	43.2%	51	40.2%	45	58.4%
Z-drug	1	7.7%	0	0.0%	9	7.1%	6	7.8%
Antiepileptic/mood stabilizer	5	38.5%	19	51.4%	53	41.7%	36	46.8%
Lithium	3	23.1%	6	16.2%	22	17.3%	8	10.4%
Unmedicated	1	7.7%	3	8.1%	14	11.0%	7	9.1%
Other med[‡*]	1	7.7%	9	24.3%	11	8.7%	13	16.9%
Past Alcohol Abuse[‖*‡*]	8	61.5%	25	67.6%	50	39.7%	55	71.4%

Note. PTSD = Posttraumatic stress disorder; Z-drug = Nonbenzodiazepines; Med = Medication.

[•] significant difference between current cannabis and past cannabis;

[†] significant difference between current cannabis and no drug use;

[‡] significant difference between past cannabis and no drug use;

[§] significant difference between current cannabis and past other drugs;

[¤] significant difference between past cannabis and past other drugs;

[‖] significant difference between no drug use and past other drugs;

[*] significant at $p < .05$